

Resin and Radiance – E-Commerce Order Management System

Entity, Attributes, Primary Keys, and Relationships

1. Customer Entity

Description

The **Customer** entity stores the personal information of customers who register on the website, browse products, place orders, and track their purchases.

Primary Key (PK)

- **Customer_ ID**

Attributes

- Customer_ ID (PK)
- Customer_ Name
- Email
- Phone_ Number
- Password
- Address
- City
- Pin code
- Registration Date

Purpose

- Stores customer information.
 - Maintains customer login credentials.
 - Keeps delivery addresses.
 - Links customers with their orders.
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2. Product Entity

Description

The **Product** entity stores information about all handmade resin products available for sale.

Primary Key (PK)

- **Product_ID**

Attributes

- Product_ID (PK)
- Product_Name
- Category
- Description
- Price
- Stock_Quantity
- Product_Image
- Availability_Status

Purpose

- Displays products to customers.
- Maintains product pricing.
- Tracks available stock.
- Supports customized resin products.

3. Orders Entity

Description

The **Orders** entity records every order placed by customers.

Primary Key (PK)

- **Order_ID**

Foreign Key (FK)

- Customer_ID

Attributes

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount
- Order_Status
- Delivery_Address
- Estimated_Delivery_Date

Purpose

- Stores customer orders.
 - Tracks order progress.
 - Calculates total purchase amount.
 - Links customer with purchased items.
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4. Order_ Item Entity

Description

The **Order_ Item** entity stores details of each product included in an order.

Primary Key (PK)

- **Order Item_ ID**

Foreign Keys (FK)

- Order_ID
- Product_ID

Attributes

- Order Item _ID (PK)
- Order_ ID (FK)
- Product_ID (FK)
- Quantity
- Unit_ Price
- Sub_ Total

Purpose

- Connects Orders and Products.
 - Stores quantity of each product.
 - Calculates subtotal.
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5. Payment Entity

Description

The **Payment** entity stores payment details for customer orders.

Primary Key (PK)

- **Payment_ ID**

Foreign Key (FK)

- Order_ ID

Attributes

- Payment_ ID (PK)
- Order_ ID (FK)
- Payment_ Method
- Payment_ Status
- Transaction_ ID
- Payment_ Date
- Amount

Purpose

- Stores payment records.
 - Tracks payment status.
 - Maintains transaction history.
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6. Shipping Entity

Description

The **Shipping** entity stores shipping and delivery information for each order.

Primary Key (PK)

- **Shipping_ ID**

Foreign Key (FK)

- Order_ ID

Attributes

- Shipping_ ID (PK)
- Order_ ID (FK)
- Courier_ Name
- Tracking_ Number
- Dispatch_ Date
- Delivery_ Date
- Delivery_ Status

Purpose

- Tracks courier details.
- Updates shipment status.
- Confirms successful delivery.

Relationship Between Entities

1. Customer → Orders

Relationship: One-to-Many (1:M)

Explanation

- One customer can place many orders.
- Every order belongs to only one customer.

Example

Customer

Customer_ID = C101

Orders

Order001

Order002

Order003

Relationship

Customer (1) -----Orders (M)

2. Orders → Order_Item

Relationship: One-to-Many (1:M)

Explanation

- One order may contain multiple products.
- Each Order_Item belongs to one order.

Example

Order001

Tray

Keychain

Bookmark

Relationship

Orders (1) ----- Order Item (M)

3. Product → Order Item

Relationship: One-to-Many (1:M)

Explanation

- One product can appear in many customer orders.
- Every Order Item refers to only one product.

Relationship

Product (1) -----Order Item (M)

4. Orders → Payment

Relationship: One-to-One (1:1)

Explanation

- Each order has one payment.
- Each payment belongs to one order.

Relationship

Orders (1) ----- Payment (1)

5. Orders → Shipping

Relationship: One-to-One (1:1)

Explanation

- Every order has one shipping record.
- Each shipping record belongs to one order.

Relationship

Orders (1) ----- Shipping (1)

Summary of Relationships

Relationship	Cardinality	Description
Customer → Orders	1: M	One customer can place many orders.
Orders → Order Item	1: M	One order can contain many products.
Product → Order Item	1: M	One product can appear in many orders.
Orders → Payment	1: 1	One order has one payment.
Orders → Shipping	1: 1	One order has one shipping record.

Primary Key Summary

Entity	Primary Key
Customer	Customer_ID
Product	Product_ID
Orders	Order_ID
Order_Item	OrderItem_ID
Payment	Payment_ID
Shipping	Shipping_ID
